Warrant

Don't let the agent grade its own homework.

别让 AI 代理给自己的作业打分。

On-chain trade limits for AI agents, a signed pre-trade disclosure, and now an independent price check: Chainlink Data Feeds re-price every disclosure on chain, and an agent that signs PASS on a bad quote is caught and loses reputation.

链上交易授权 + 交易前签名披露 + 新增独立价格复核：Chainlink 在链上重算每一份披露，对坏报价签 PASS 的代理会被当场揭穿并扣信誉。

An AI agent's trading limits, on chain and checkable.

把 AI 代理的交易边界放上链，让任何人都能核查。

01
Grant
授权人设定单笔上限、滑点、价格冲击、费用等，可随时撤销。

`WarrantRegistry`

02
Preflight
代理下单前，按链上授权给真实报价打分。

`mm warrant preflight`

03
Disclose
代理钱包签名打分结论，任何人可代为上链。

`EIP-712 → WarrantAttestor`

04
Reputation
每份已验证的披露写成 ERC-8004 信誉记录。

`WarrantReputation`

05 · NEW
Price check
Chainlink 独立重算报价，核对代理说的是不是真话。

`WarrantPriceCheck`

Arc
mainnet, chain 5042
已上主网

ERC-8004
Arc's own registries, agent 1419
接入 Arc 官方注册表

56 + 65
contract + plugin tests, all passing
合约与插件测试全部通过

MIT
open source, every tx linked in README
开源，每笔交易可追溯

Steps 01–04 were built before this event (Monad Metropolis, Avalanche Buildathon, Arc Microgrants); step 05 is the increment for ETH Shenzhen 2026. 01 – 04 为既有工作，05 为本届新增。

Every number on the scorecard came from the agent.

打分卡上的每个数字，原本都由代理自己算。

```
// what the agent signs (EIP-712)
PreflightAttestation {
  principal      0x3fd7…0f73
  agent          0x299e…0e0e
  quoteHash      keccak(quote id)
  warrantVersion 1
  verdict        PASS
  scorecardHash  0x…
}

// checked on chain: signer = agent,
// warrant active at that version.
// NOT checked: was PASS true?
```

A modified plugin can print PASS on anything.

插件被改过，任何报价都能显示 PASS。

An agent can skip preflight and sign PASS anyway.

代理可以不打分，直接签 PASS。

A bad quote can come with flattering price data.

坏报价也可能附带好看的价格数据。

Until now, the chain could not tell.

此前，链上无从分辨。

Source: `contracts/src/WarrantAttestor.sol` (`submit` verifies the signer and the warrant version only).

Chainlink re-prices it. The contract checks the verdict.

Chainlink 重算报价，合约核对代理的结论。



AGENT DISCLOSED	WITHIN BUDGET	OVER BUDGET, INSIDE ORACLE TOLERANCE	OUT OF BOUNDS
PASS 报通过	**consistent +1**	**consistent +1** (plugin warns)	**CONTRADICTED −1** 说谎
WARN / FAIL 报警告/不通过	**consistent +1**	**consistent +1**	**consistent +1** (it told the truth)

Stale feed (> 25 h) or non-positive price → revert: an unpriceable quote is never recorded as checked. Tolerance 100 bps = the two feeds' 0.5% deviation thresholds. No admin; feeds fixed at deployment. 价格源过期即回滚，不记录。

A real quote, a real agent wallet, two Chainlink feeds.

真实报价、真实代理钱包、两个 Chainlink 价格源。

0 bps

shortfall: Chainlink priced 0.5 USDC at 448,542 EURC units; the quote offered 449,499.

报价比 Chainlink 公允值还高，价格复核通过；总结论仍是 FAIL（gas 占 199 bps，上限 50）。

STEP	TRANSACTION	RESULT
Deploy WarrantPriceCheck 1,258,254 gas	`0xcc34…b956`	`status 1`
mm wallet signs FAIL, relayed + rated real `mm swap quote` 0xc700…1a29	`0x9c4a…02be`	`preflight / fail`
checkAndRate fair 448,542 · quoted 449,499	`0xfcce…4117`	`consistent`
Staged dishonest agent signs PASS quote 5% below Chainlink-fair	`0xbde8…8f4d`	`contradicted`

Source: receipts and `PriceChecked` events read from rpc.mainnet.arc.io, 2026-10-14. Feeds: USDC/USD `0x84EA…7905`, EURC/USD `0x361b…Bf23`. The dishonest agent is staged and labelled so in its ERC-8004 registration. No swap was executed.

Same bad quote. One agent tells the truth.

同一笔坏报价，一个代理说真话，一个说谎。

Honest agent 诚实代理

```
$ mm warrant preflight --price-check …
fair     448542 EURC units
quoted   426114
X Chainlink price check
  500 bps below fair · limit 200 (+100)

signs FAIL → relayed → checkAndRate
PriceChecked
  out of bounds   true
  contradicted    false

ERC-8004  consistent 1 · contradicted 0
```

Dishonest agent 说谎代理

```
$ # skips preflight
fair     448542 EURC units
quoted   426114
  (never looked)


signs PASS → relayed → checkAndRate
PriceChecked
  out of bounds   true
  contradicted    true

ERC-8004  consistent 0 · contradicted 1
```

Output of `node scripts/demo-pricecheck.mjs`: an Arc mainnet fork with the deployed contracts and live feeds; both agents and the quote are staged. 本地分叉 Arc 主网运行，零成本，可现场复现。

Reveal, not enforce. Now with a second opinion.

只揭示，不拦截；现在多了一个独立的第二意见。

Only tokens with a feed are checkable (on Arc: USDC, EURC).
只能复核有 Chainlink 价格源的币种。

A 100 bps tolerance absorbs oracle lag; smaller lies pass.
100 bps 容差吸收价格源滞后，更小的偏差不会被标记。

The check runs when someone calls it; unchecked is visible, not penalised.
有人调用才复核；未复核可见，但不扣分。

Gas outside the quote is not in the output; the fee-load check covers it.
报价外的 gas 不计入，由费用占比一项覆盖。

Mainnet, not audited; the contracts hold no funds.
已上主网，未审计；合约不托管资金。

CODE · 代码

github.com/Beltran12138/warrant

RUN THE DEMO · 复现演示

```
npm install && npm run build
node scripts/demo-pricecheck.mjs
```

BUILT FOR THIS EVENT · 本届新增

WarrantPriceCheck, 23 contract tests, 6 plugin tests, `--price-check`, relayer step, Arc mainnet deployment.
独立开发，使用 AI 编程工具，已在 README 披露。

Chainlink usage, files and transactions: README › Chainlink price check on Arc mainnet. Chainlink 用到的文件与交易清单见 README。